PEREKA 2026 · BRIEFING NOTES

The wow factor — and how MathXplore compares

What makes the lab memorable in a 60-second visit, what else exists, and how to talk about the difference honestly.

Prepared 24 September 2026. Every link was checked that day; claims about MathXplore were checked against its source code.

THE SHORT ANSWER

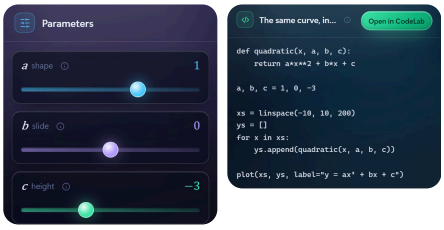
MathXplore treats maths, pictures and code as one thing.

Most tools do one of these well. MathXplore makes you move between all three, predict before you see, and learn from the exact mistakes maths students make — and it keeps working with the wifi off.

Four wow moments — 60 seconds in front of a judge

01 Drag a slider

Explore → Quadratic Functions

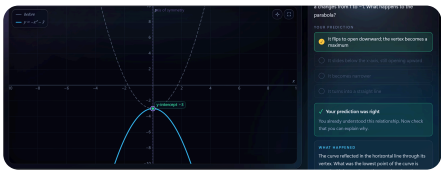


The parabola moves; its vertex, roots, axis of symmetry and discriminant update live. The Python beside it changes too: `a, b, c = 1, 0, -3` becomes `a, b, c = -1, 0, -3`. That panel is generated from the slider values (`src/pages/explore/QuadraticModule.tsx`, line 111), and *Open in CodeLab* carries it over to run and edit.

Why it lands: the judge sees in one gesture that the equation, the picture and the program are the same object.

02 "What happens if a goes from 1 to −1?"

The What If? panel on the same page, scenario 1

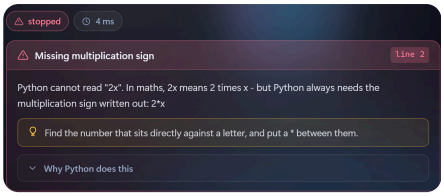


The judge has to commit to an answer — the button reads *Choose a prediction first* until they do. Then *Lock in and see what happens*: the curve animates against a ghost of the old one, and the lab explains what happened, the concept behind it, and the same change in Python.

Why it lands: committing to an answer makes them care about the reveal — and they find out whether their own intuition was right.

03 Type `y = 2x + 3`

CodeLab



Most tools answer with `SyntaxError: invalid syntax`. MathXplore says *"Missing multiplication sign — Python cannot read "2x". In maths, 2x means 2 times x, but Python always needs the multiplication sign written out: 2*x"*, then gives a hint that stops short of the answer, and a "Why Python does this" note.

The engine has lessons like this for other maths habits too: `x^2` instead of `x**2` · `=` versus `==` · square roots of negative numbers · dividing by zero · a function that never returns its answer · counting from 0.

Why it lands: every maths student makes this mistake, and here the mistake becomes the lesson.

04 Switch the laptop to airplane mode

Anywhere in the app



Everything keeps working — navigate, rotate the 3D views, run code — with no account and nothing sent anywhere. The Python engine is part of the page, so there is no download to wait for.

Why it lands: at a Sabah event, where school connectivity is a real problem, this is the moment that turns "nice" into "we could use this".

SET-UP TIP

Run it from the laptop itself (`npm run preview`, a *localhost* address). Airplane mode doesn't affect localhost. A hosted copy would fail on a full page reload with no connection.

· THE LANDSCAPE

How it compares

Names are links. Rows are categories of tools; examples are the best-known ones.

EXISTING TOOL	WHAT IT DOES WELL	WHAT MATHXPLORE ADDS
Graphing calculators Desmos · Desmos 3D GeoGebra · offline apps	Excellent, free 2D and 3D graphing with sliders, geometry and algebra.	Python tied to the same picture, predict-then-reveal questions, errors explained in maths terms, challenges graded by running code.
GeoGebra's Python app CLOSEST IDEA PyGgb web app · docs	"Allows users to create GeoGebra constructions using the Python programming language" — code that draws maths.	A guided loop around the code — lessons, What If? predictions, maths-aware error lessons, graded challenges — running offline from files you keep.
Brilliant brilliant.org	Polished, interactive "try it first" lessons in maths and coding.	Free; the maths and the code on the same screen as one object rather than in separate courses; works offline.
Jupyter notebooks with slider widgets Jupyter · ipywidgets · JupyterLite	Real Python, sliders and plots. JupyterLite runs in the browser using Pyodide.	No set-up and no large first download; errors explained for a maths learner; a structured path instead of a blank notebook.
Online Python editors Replit · Programiz Trinket SHUT DOWN	Real Python in a browser tab, some with turtle graphics.	Works without a connection or an account; errors explained; every program linked to the maths it draws.
Beginner-friendly Python Thonny · Python Tutor Futurecoder · friendly-traceback	Friendlier error messages and step-by-step views of running code.	Errors explained in *maths* terms, linked to notation the student already knows; graphs, shapes and 3D surfaces.
Python graphing calculators NumWorks · TI-84 Plus CE Python	Python on a device built for maths.	No hardware to buy; code, sliders and graphs linked on one screen; explained errors.

Trinket, a popular classroom Python editor, shut down on 31 August 2026; its saved trinkets, courses and assignments are no longer accessible. It's a useful example of the risk of depending on a hosted service — MathXplore runs from static files the school keeps.

• BEING HONEST ABOUT IT

What’s genuinely different

Every ingredient exists somewhere. What's new is **the combination, in one page that runs anywhere**. Four parts of it are hard to copy:

Errors explained in maths terms

Friendlier Python errors exist (Thonny, Futurecoder, friendly-traceback), but they explain Python in Python's terms. MathXplore's lessons start from the maths habit — "$2x$ means 2 times x", "^ isn't a power" — because the team wrote its own Python engine (about 3,900 lines of TypeScript) and owns every message.

Answers judged by running them

Challenges execute the student's function at test points, so any correct solution passes — including one nobody anticipated. The student's own function is drawn over the goal, so a wrong answer looks exactly as wrong as it is.

One loop around everything

Learn → explore → predict → change → visualise → code → test → understand → create, lit up across the top of every working screen, with the maths, the picture and the Python on the same page.

Zero set-up, zero data

Loads instantly with no login, keeps every record on the device, and costs almost nothing to host because it is just static files. No multi-megabyte Python download, as a Pyodide-based tool needs.

DON’T SAY

- "The first …" or "the only …"
- "Improves results by …" — nothing has been measured yet
- "Full Python"

SAY INSTEAD

- "MathXplore **brings these together in one place**."
- "It's **evidence-ready**: the logging for a pre/post-test study is built in."
- "A deliberate **teaching subset** of Python."

A judge who knows one counter-example to "the first" will discount everything else you say.

ONE-LINE PITCH

”Desmos shows you the maths, Python editors teach you the code. MathXplore is where they meet — and it works with the wifi off.”

· AT THE BOOTH

Where judges will push

"Desmos is free and polished. Why use yours?"	Desmos teaches the picture, and Python editors teach the code. MathXplore is the bridge between them — with prediction and maths-aware errors — and it works offline.
"GeoGebra already lets you write Python."	Yes — PyGgb is the closest idea, and it shows the direction matters. It's an open coding tool; MathXplore wraps the code in a guided loop of lessons, What If? predictions, explained errors and graded challenges, and runs from files the school keeps.
"Does it work?"	It hasn't been measured yet, and we don't claim it. The event log, CSV/JSON export and anonymous participant codes are built; the next step is a school pilot with a pre- and post-test.
"Why is there so little content?"	6 lessons and 12 challenges is small next to Khan Academy or Brilliant. They prove the method works; teacher authoring tools are on the roadmap.
"Is it real Python?"	A deliberate teaching subset: variables, loops, functions, lists, the maths module and plotting — no classes or dictionaries. Its behaviour is tested against real Python (119 engine checks).

UPDATED SINCE OUR CHAT

- **GeoGebra now has a Python app (PyGgb)** that builds GeoGebra constructions from Python code. The chat version said graphing apps had "no path to Python"; that was wrong, and PyGgb is now listed as the closest idea.
- **Trinket shut down on 31 August 2026.** It's kept in the table, marked as such, with Programiz added as a working example.
- **Friendlier error messages exist elsewhere** (Thonny, Futurecoder, friendly-traceback). The chat said a Pyodide-based tool "can't rewrite the error messages"; it can. MathXplore's real difference is explaining errors in maths terms, with no download.
- **"The first place these come together"** contradicted the "don't say first" advice; it's now "brings these together in one place".
- Casio's Python calculators were dropped from the table: their site blocks automated checks, so the link couldn't be verified.

• REFERENCE

All links

Checked 24 September 2026. Each opened successfully (HTTP 200) except where noted.

Desmos Graphing Calculator
https://www.desmos.com/calculator

Desmos 3D Calculator
https://www.desmos.com/3d

GeoGebra
https://www.geogebra.org

GeoGebra offline apps
https://www.geogebra.org/download

GeoGebra Python (PyGgb) web app
https://www.geogebra.org/python/index.html

PyGgb documentation
https://www.geogebra.org/python/doc/index.html

Brilliant
https://brilliant.org

Jupyter
https://jupyter.org

ipywidgets
https://ipywidgets.readthedocs.io

JupyterLite
https://jupyterlite.readthedocs.io

Pyodide
https://pyodide.org

Replit
https://replit.com

Programiz online Python compiler
https://www.programiz.com/python-programming/online-compiler/

Trinket
shut down 31 Aug 2026; the page now carries the notice
https://trinket.io

Thonny
https://thonny.org

Python Tutor
https://pythontutor.com

Futurecoder
https://futurecoder.io

friendly-traceback
https://friendly-traceback.github.io/docs/

NumWorks
https://www.numworks.com

TI-84 Plus CE Python
https://education.ti.com/en/products/calculators/graphing-calculators/ti-84-plus-ce-python

Khan Academy
https://www.khanacademy.org